Supplementary Table 2. Study Characteristics and Risk of Bias Assessment

Study (Author, Year, Country)	Study Design & Objective	Sample Characteristics	TBI and MHSU Information		Integration Information				WHO Strategic Approach	Facilitator	Barrier	Risk of Bias Assessment
			TBI Severity/Time Since Injury (Mean ± SD, Years, Unless Specified)	Type of MHSU/Time Since Dx (Mean ± SD, Years, Unless Specified)	Activity	Description	Level	Type				
Ahmed et al., 2016 United States	Case Study Integrated model of care to address physical, cognitive, and psychiatric needs among individuals w/TBI	N=1 - Gender: Man - Age [Year]: 31 at time of tx; 24 at time of TBI - Marital Status: Divorced - Education: GED	Severe/7 years	ADHD/ Elementary school; MDD/NR	Program	Program of Enhanced Psychiatric Services (PEP) (Outpatient psychiatric brain injury clinic) - Individual and gp psychotherapy + pharmacotherapy - Weekly multidisciplinary team meetings - Individualized tx plans - Family meetings	Meso	Service	1.1 1.4	- Integrating family members - Staff experienced in managing behavioural sxs from TBI	- Geographic distance b/t PEP and neurorehabilitation outpatient program	- Selection - Reporting - Information
Albrecht et al., 2017 United States	Qualitative Explore perception of barriers and facilitators to dx and tx of MHSU after TBI	N=33 (10 HCP; 18 patients; 5 caregivers) - Gender [Men]: 83% Patients; 40% Caregivers; HCP NR	Mild-Severe/ 18 mos-5 years	Depression; Anxiety; PTSD/NR	Screening, Treatment	Dx and tx of MHSU after TBI	Meso	Clinical	4	- Positive and engaged caregiver support	- Lack of education about TBI and MHSU among patients, HCP, and caregivers - Lack of experience with TBI and MHSU among HCP --> reluctance to dx, which delays tx - polypharmacy --> complicates tx of TBI and MHSU - Limited access to care due to insurance, transportation, income, and availability of providers	N/A
Andrew et al., 2017 South Africa	Cohort Incorporate questions re: cognition and MHSU sxs to TBI screening tool to identify patients for referral	N=47 - Sex [Males]: 94% - Age [Years]: 16 to 75; mean 35 - Employment [Unemployed]: 36% Pre-injury; 11% at time of study	Mild/moderate (N=11); Severe (N=32); Unknown (N=4)/NR	Depression; Anxiety; Hallucinations/NR	Screening	Screening for MHSU among individuals with TBI to improve referral and patient outcome	Meso	Clinical	4	- Interpreter and family/caregiver support	- Communication (due to language and education) --> lack of understanding of some questions - Cultural differences --> difficulty defining depression and personality change	N/A

Ansari et al., 2014 India	RCT Evaluate role of sertraline in depression and impact on QoL after TBI	N=80 - Sex [Males]: 100% - Age [Years]: 18-24 (35%); 25-34 (27.5%); 35-44 (16.3%); 45-54 (15%); 55-64 (5%); ≥65 (1.2%)	Mild (54%); Moderate (46%)/ ≥2 weeks - <3 mos (30%); 3-6 mos (24%); >6 mos (46%)	Depression/NR	Treatment	Pharmacotherapy (50mg Sertraline/day) over 6 mos	Micro	Clinical	1.1	N/A	N/A	- Selection - Performance - Non-differential misclassification - Information
Ashish et al., 2017 United States	Case Study Describe patient in outpatient behavioural health clinic post-TBI	N=1 - Sex: Male - Age [Years]: 14 - Ethnicity: Hispanic	Mild/7 weeks (2 additional concussions w/in past 2 years)	Anxiety/NR; Depression/NR; Hallucination/3 weeks post-TBI	Treatment	Individual and family psychotherapy (Acceptance and Commitment Therapy) + vestibular-ocular therapy	Micro	Service	1.1 4.1	N/A	N/A	- Reporting - Information
Ashman et al., 2014 United States	RCT Efficacy of CBT vs. SPT in treating depression among individuals w/TBI	N=54 (28 CBT; 26 SPT) CBT: - Sex [Males]: 35.7% - Age [Mean ± SD, Years]: 47.5 ± 11 - Race/Ethnicity: White (71.4%); Black/African American (10.7%); Hispanic/Latino (10.7%); Other (7.2%) - Marital Status: Single (50%); Married/Live Together (25%); Widowed/Divorced/Separated (25%) - Education: ≤High School/GED (17.9%); Trade School/Some College (21.4%); Bachelor's Degree (21.4%); ≥Master's Degree (39.3%) - Income: <$10,000 (25.9%); $10,001–$20,000 (22.2%); $20,001–$60,000 (29.7%); >$60,000 (22.2%) SPT: - Sex [Males]: 46.1% - Age [Mean ± SD, Years]: 47.0 ± 10.8 - Race/Ethnicity: White (46.1%); Black/African American (15.4%); Hispanic/Latino (23.1%); Other (15.4%) - Marital Status: Single (46.1%); Married/Live Together (23.1%); Widowed/Divorced/Separated (30.8%) - Education: ≤High School/GED (23.1%); Trade School/Some College (42.3%); Bachelor's Degree (11.5%); ≥Master's Degree (23.1%) - Income: <$10,000 (23.1%); $10,001–$20,000 (15.4%); $20,001–$60,000 (38.4%); >$60,000 (23.1%)	CBT: Mild (37%); Moderate-Severe (63%)/7.8 ± 13.4 SPT: Mild (43%); Moderate-Severe (57%)/ 13.2 ± 18.3	Depression/NR	Treatment	Psychotherapy (CBT vs. SPT) - CBT adapted to include accommodations for cognitive challenges	Micro	Clinical	1.1	N/A	N/A	- Small sample - Attrition - Selection

Baig et al., 2019 United States	Case Control Describe model of integrated psychiatric care for PTSD and mTBI	N=100 (50 UC, 50 IC) UC: - Age [Mean ± SD, Years]: 33.7 ± 8.24 - Gender [Men]: 98% - Marial Status [Not Married]: 25% - Education: HS Diploma (44%); Some College (48%); 4-Year College Degree (8%) - Employment [Unemployed]: 62% - Race: Hispanic/Latino (32%); Caucasian (52%); African American (16%) IC: - Age [Mean ± SD, Years]: 34.9 ± 7.8 (IC) - Gender [Men]: 98% - Marital Status [Not Married]: 20% - Education: HS Diploma (36%); Some College (48%); 4-Year College Degree (16%) - Employment [Unemployed]: 50% - Race: Hispanic/Latino (36%); Caucasian (48%); African American (16%)	Mild/NR	PTSD/ 2.6 ± 2 (UC); 1.75 ± 2.3 (IC)	Program	UC: - SW conduct behavioural health evaluation outside of facility - Psychotherapy (most commonly CPT) provided by psychologist at MH outpatient service - Pharmacotherapy, if needed, provided by psychiatrist/physician assistant/advanced practice registered nurse specialist in psychiatry/clinical pharmacy specialist at MH outpatient service - Communications via EMR IC: - Psychologist and psychiatrist embedded w/in team at facility - Pharmacotherapy provided by neuropsychiatrist - Psychotherapy (most commonly individual CPT) provided by behavioural therapist - Multidisciplinary team meetings - Alerts for providers via EMR	Meso	Service Functio nal	1.1 4.1	- Integration and collaboration of MHSU care in small, co-locating, and focused polytrauma rehabilitation setting --> improved access to and engagement in tx - Collaboration of psychiatrist and behavioural therapist --> outcome success	N/A	- Referral - Selection - Reporting - Performance
Baig et al., 2019 United States	Case Series Evaluate outcomes of Valproate, Risperidone, and Quetiapine on completion of CPT for PTSD among individuals with TBI	N=50 (8 Valproate; 17 Risperidone; 25 Quetiapine) Valproate: - Gender [Men]: 100% - Age [Mean ± SD, Years]: 37 ± 7 - Ethnicity [Hispanic]: 38% - Race: White (100%) - Marital Status [Not Married]: 38% - Education: GED (13%); HS Diploma (63%); College Degree (25%) - Employment [Unemployed]: 50% Risperidone: - Gender [Men]: 94% - Age [Mean ± SD, Years]: 36 ± 10 - Ethnicity [Hispanic]: 35% - Race: Black (24%); White (76%) - Marital Status [Not Married]: 82% Risperidone (82%) - Education: GED (18%); HS Diploma (29%); Some College (29%); College Degree (24%) - Employment [Unemployed]: 65% Quetiapine: - Gender [Men]: 96% - Age [Mean ± SD, Years]: 37 ± 8 - Ethnicity [Hispanic]: 32% - Race: Asian (8%); Black (20%); White (72%) - Marital Status [Not Married]: 72% - Education: GED (8%); HS Diploma (52%); Some College (28%); College Degree (12%) - Employment [Unemployed]: 68%	mTBI/NR	PTSD/ 2 ± 3 (Valproate); 2 ± 2 (Risperidone); 3 ± 8 (Quetiapine)	Treatment	Psychotherapy (CPT and prolonged exposure therapy) + pharmacotherapy (Valproate, Risperidone, Quetiapine)	Meso	Service	1.1	N/A	N/A	- Selection - Information - Attrition

Bay et al., 2017 United States	Case Study Case management model for managing chronic stress and depression after TBI	N=1 - Gender: Man - Age [Years]: 90 - Employment: Retired - Education: "Well-educated" - Martial Status: Widowed (following 73 year relationship/marriage)	NR/NR	Depression/Prior to TBI	Treatment	Case management with caregiver: - Case manager plans for natural support system, family education, crisis intervention for maintaining QOL; recommends sxs management, reducing loneliness and depression, limiting health service use - Case manager with family to make changes at home and activities for within and outside of home	Micro	Clinical	1.1 4.1	- Knowledge of evidence-based practice, protocols, and pathways --> case manager communicate effectively w/interdisciplinary professional and others to engage in proactive patient advocacy	N/A	- Selection - Reporting - Information
Bédard et al., 2014 Canada	RCT Examine efficacy of MBCT for depression among individuals w/TBI	N=76 (38 Tx; 38 Control) Tx: - Sex [Males]: 50% - Age [Mean ± SD, Years]: 47.1 ± 12.0 - Marital Status: Married/Common law (58%); Single (26%); Separated/Divorced/Widowed (13%); Unknown (3%) - Employment: Employed (37%); Unemployed (21%); Homemaker/Volunteer (11%); Retired (11%); Other (21%) - Education: Elementary (3%); Some Secondary (11%); Completed Secondary (32%); Some Post-Secondary (11%); Completed Post-Secondary (45%) Control: - Sex [Males]: 60% - Age [Mean ± SD, Years]: 45.8 ± 14.8 - Marital Status: Married/Common law (45%); Single (31%); Separated/Divorced/Widowed (24%); Unknown (0%) - Employment: Employed (24%); Unemployed (26%); Homemaker/Volunteer (0%); Retired (16%); Other (34%) - Education: Elementary (5%); Some Secondary (11%); Completed Secondary (18%); Some Post-Secondary (16%); Completed Post-Secondary (50%)	Tx: NR/4.5 ± 4.1 Control: NR/4.0 ± 3.5	Depression/NR	Treatment	MBCT: - customized to address issues associated with TBI	Micro	Clinical	1.1	N/A	N/A	- Nondifferential misclassification - Selection - Attrition - Performance
Bombardier et al., 2017 United States	RCT Explore relationship between depression severity and cognition, behaviour, and physical activity among individuals w/TBI receiving tx for depression	N=100 (40 CBT-T; 18 CBT-IP; 42 UC) CBT: - Sex [Males]: 59% - Age [Mean ± SD, Years]: 45.4 ± 14.1 - Race/ Ethnicity: Non-Hispanic White (90%) - Marital Status: Single/Never Married (31%); Married/Partnered (19%); Divorced (41%); Separated (5%); Widowed (2%); Other (2%) - Education: Some College (45%); Undergrad Degree (16%); Grad Degree (5%); Tech/Vocational (9%); High School Diploma (14%); ≤High School/GED (12%) UC: - Sex [Males]: 69% - Age [Mean ± SD, Years]: 46.3 ± 12.4 - Race/ Ethnicity: Non-Hispanic White (90%) - Marital Status: Single (29%); Married/Partnered (31%); Divorced (36%); Separated (5%) - Education: Some College (45%); Undergrad Degree (17%); Grad Degree (17%); Tech/Vocational (5%); High School Diploma (5%); ≤High School/GED (12%)	CBT: "complicated, mild/moderate" (69%); Severe (31%)/3.41 ± 2.84 UC: "complicated, mild/moderate" (69%); Severe (31%)/3.21 ± 2.58	Depression/NR Other MHSU: Current Dysthymic (12%); Alcohol Abuse (10%); Substance Abuse (4%); hx of PTSD (18%)/NR	Treatment	CBT-T and CBT-IP - adaptations to make protocol more accessible for people with cognitive impairment UC: - Notified by phone of depression status and encouraged to continue using rehabilitation and primary care services - Free to self-refer to MHSU services outside of study - Local MH resources provided to patient	Micro	Clinical	1.1	N/A	N/A	- Selection - Nondifferential misclassification - Ascertainment - Recall

Boyd et al., 2016 United States	Case Study Determine efficacy of CPT in tx of PTSD among individuals with hx of mTBI	N=1 - Sex: Male - Age [Years]: "early 40s" - Race/Ethnicity: Caucasian - Marital Status: "Separated from wife" - Education: High School + 1 year of College	mTBI/1 year prior to tx	PTSD/10+ years, sought care 5 mos post-TBI	Treatment	CPT-Cognitive Only	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Performance
Brenner et al., 2015 United States	Cohort Screening for TBI among veterans seeking MH services and establish classification accuracy of TBI-4 using OSU-TBI as criterion standard	N= 316 (238 OSU TBI-ID pos; 78 OSU TBI-ID neg) OSU TBI-ID pos: - Sex [Males]: 92% - Race/ Ethnicity: White (54%); African American (25%); Hispanic (13%); Other (7%) OSU TBI-ID neg: - Sex [Males]: 85% - Race/ Ethnicity: White (62%); African American (22%); Hispanic (8%); Other (9%)	Severity of worst TBI: Altered consciousness (22%); <5 min (35%); 5-30 min (16%); >30min (27%) Time since first TBI: 33.2 ± 15.2	NR (all patients sought MH services)/NR	Screening	Screening for TBI using TBI-4 in MHSU setting	Meso	Clinical	4	N/A	N/A	N/A
Brown et al., 2019 United States	Cohort Examine efficacy of neurofeedback therapy among IPV survivors w/TBI	N=32 - Sex [Males]: 3% - Age [Mean, Years]: 46.9 - Ethnicity: White (63%); Hispanic (19%); Black/African American (13%); Asian (5%)	Moderate/NR	Depression; Anxiety; PTSD/NR	Treatment	Neurofeedback therapy	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Performance
Carrick et al., 2015 United States	Before-After, No Control Evaluate effectiveness of vestibular rehabilitation to reduce PTSD sxs among individuals w/combat-related TBI	N=98 - Sex [Males]: 100% - Age [Mean, Years]: 39	NR/NR	PTSD/NR	Treatment	Vestibular rehabilitation	Micro	Clinical	1.1	N/A	N/A	- Selection - Performance
Carrick et al., 2015 United States	Before-After, No Control Understand LT efficacy of vestibular rehabilitation in reducing PTSD sxs among individuals w/combat-related TBI	N=26 - Sex [Males]: 100% - Age [Mean, Years]: 38.5	NR/NR	PTSD/NR	Treatment	Vestibular rehabilitation	Micro	Clinical	1.1	N/A	N/A	- Selection - Performance

Cole et al., 2015 United States	Before-After, No Control Explore safety, acceptability, feasibility, and efficacy of Mindfulness-Based Stress Reduction on neurocognitive and psychiatric measures in individuals w/PTSD and hx of mTBI	N=9 - Age [Mean ± SD, Years]: 45.6 ± 11.6 -Race/Ethnicity: White/ Non-Hispanic (33.3%); Hispanic/Latino (22.2%); African American (44.4%)	mTBI/ ≥12 months	PTSD/NR	Treatment	Mindfulness-Based Stress Reduction	Micro	Clinical	1.1	- Appointment reminder	N/A	- Information - Selection - Performance
D'Antonio et al., 2013 United States	RCT Examine differential tx effects of CBT and SPT on self-reported sxs of depression in individuals w/TBI who sought tx for depression	N=44 (22 CBT; 22 SPT) - Sex [Females]: 57.1% - Age [Mean ± SD, Years]: 48.8 ± 10.2 - Race/Ethnicity: Caucasian (57%); Hispanic (20%); African American (15%) - Education: ≤ High School (27%); Some College/College Degree (36%); Beyond College Degree (23%)	Mild (9%); Moderate (9.3%); Severe (15%)/7.69 (SD NR)	MDD/NR	Treatment	CBT: - Modified to address cognitive deficits associated with TBI SPT	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Ascertainment
Dindo et al., 2020 United States	Qualitative and RCT Develop and refine ACT workshop and examine acceptability, feasibility and effect of ACT for veterans w/mTBI and MHSU	**Phase 1** N= 11 Veterans (Workshop 1 = 5; Workshop 2 = 6) - Gender [Men]: 100% - Age [Mean ± SD, Years]: 37.5 ± 6.1 - Ethnicity: Hispanic (45%); African American (36%); White (18%) - Marital Status: Currently Married (45%) - Education [Mean ± SD, Years]: 14.3 ± 2.1 - Employment [Currently Employed]: (45%) **Phase 2** N= 32 (20 ACT; 12 UC) ACT - Age [Mean ± SD, Years]: 37.7 ± 6.3 - Ethnicity: White (42%); African American (21%); Hispanic/Latino (21%); Other (15%) - Marital Status: 68% - Education [Mean ± SD, Years]: 14.1 ± 1.8 - Employment: 47% UC - Age [Mean ± SD, Years]: 34.7 ± 5.8 - Ethnicity: White (42%); African American (17%); Hispanic/Latino (25%); Other (17%) - Marital Status: 50% - Education [Mean ± SD, Years]: 14.3 ± 1.6 - Employment: 58%	mTBI/NR (Phase I & II)	PTSD; Substance Dependence/NR	Treatment	Acceptance and Commitment Therapy - Modified for veterans to include more veteran-centred language and experiences - Clinical psychologist, neuropsychiatrist, cognitive psychologist, chaplain and anthropologist provided feedback on exercises and examples in development of protocol	Meso	Service	1.1 4.1	- Incorporating veteran into tx team - Greater use of military terminology - Increased time reviewing manuals and practicing learned skills during session - Adjusting workshop timing	- Cognitive barriers to using patient manual and workshop skills (e.g., forgetting learned skills, misplacing manual)	- Selection - Information - Ascertainment
Eads et al., 2015 United States	Case Study Alert hypnosis for PTSD sxs	N=1 - Sex: Male - Marital Status: "Twice divorced" - Education: "Minimal education"	NR/NR	PTSD/NR	Treatment	Alert Hypnotic Induction	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Performance - Reporting

Elbogen et al., 2019 United States	RCT Test effects of cognitive rehabilitation on executive function and emotion regulation in veterans w/TBI and PTSD	N=112 (55 Control; 57 CALM) Control: - Age [Mean ± SD, Years]: 36.3 ± 8.3 - Sex [Males]: 90% - Race [Minority Status]: 50% CALM: - Age [Mean ± SD, Years]: 36.8 ± 8.6 - Sex [Males]: 90% - Race [Minority Status]: 34%	Control: Moderate/Severe (53%)/NR CALM: Moderate/Severe (61%)/NR	PTSD/NR	Treatment	Cognitive Rehabilitation (Cognitive Applications for Life Management) - Included mobile app to break down goals into steps and to record steps and set reminders to complete action items Control: - Psychoeducation on TBI - Mobile device to train visual memory	Micro	Clinical	1.1 3.5	N/A	N/A	- Selection - Information
Elbogen et al., 2019 United States	Before-After, No Control Examine feasibility of mobile neurofeedback therapy to manage pain in veterans w/TBI and PTSD	N= 36 - Age [Mean ± SD, Years]: 38.6 ± 10.0 - Sex [Males]: 85.4% - Race: White (41.5%); Black (39.0%); American Indian (4.9%); Other (14.6%) - Education [Mean ± SD, Years]: 13.6 ± 4.4	No LOC (30.8%); <1min LOC (20.5%); 1-15 min LOC (33.3%); 16-30 min LOC (7.7%); >30 min LOC (7.7%) / 13.1 ± 9.9	PTSD/NR	Treatment	Neurofeedback therapy - Portable headset linked to an application on mobile device	Micro	Clinical	1.1	N/A	N/A	- Information - Performance
Fann et al., 2015 United States	RCT Test efficacy of CBT to tx depression w/in 10 years of complicated mild to severe TBI	N=100 (40 CBT-T; 18 CBT-IT; 42 UC) CBT: - Sex [Males]: 59% - Age [Mean ± SD, Years]: 45.4 ± 14.1 - Race: Non-Hispanic White (90%) - Education: Some College (53%); College Degree (21%); High School Diploma (14%); ≤GED (12%) - Marital Status: Single (31%); Married/Partnered (19%); Divorced (41%); Separated (5%); Widowed (2%); Other (2%) UC: - Sex [Males]: 69% - Age [Mean ± SD, Years]: 46.3 ± 12.4 - Race: Non-Hispanic White (90%) - Education: Some College (50%); College Degree (33%); High School Diploma (5%); ≤GED (12%) - Marital Status: Single (29%); Married/Partnered (31%); Divorced (36%); Separated (5%)	CBT: "Complicated mild/moderate" (69%); Severe (31%)/3.41 ± 2.84 UC: "Complicated mild/moderate" (69%); Severe (31%)/ 3.21 ± 2.58	Depression/NR	Treatment	CBT-T and CBT-IP: - tailored to individuals with TBI UC: - Notified by phone of depression status and encouraged to continue using rehabilitation and primary care services available - List of local MH and TBI resources provided to self-refer to MH services outside of study	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Performance

Fann et al., 2017 United States	RCT Compare efficacy of pharmacotherapy for depression w/in 1 year of complicated mild to severe TBI	N=62 (31 Sertraline; 31 Placebo) Sertraline: - Age [Mean ± SD, Years]: 38.0 ± 12.3 - Sex [Males]: 74% - Race: Non-Hispanic White (81%); Hispanic/Latino (6%); Non-Hispanic Black (6%); Asian/Pacific Islander (3%); Other (3%) - Education [Completed High School]: 77% - Marital Status: Never Married (35%); Married (35%); Divorced/Separated/Widowed (29%) Placebo: - Age [Mean ± SD, Years]: 36.9 ± 12.9 - Sex [Males]: 77% - Race: Non-Hispanic White (81%); Hispanic/Latino (6%); Non-Hispanic Black (10%); Other (3%) - Education [Completed High School]: 74% - Marital Status: Never Married (45%); Married (19%); Divorced/Separated/Widowed (35%)	Sertraline: Complicated Mild (52%); Moderate (19%)/NR Placebo: Complicated Mild (42%); Moderate (23%)/NR	Depression/NR	Treatment	Psychotherapy (Sertraline)	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Attrition
Gertler et al., 2019 Australia	Case Study Investigate Behavioural activation therapy to improve activity participation and mood for individuals with depression after TBI	N=2 - Age [Years]: 26, 46 - Gender [Men]: 100% - Marital Status: Separated - Employment: Unemployed	Mild/2 years Severe/1 year 3 mos	MDD (N=2); Alcohol Dependence (N=1)/NR	Treatment	Behavioural Activation Therapy	Micro	Clinical	1.1 3.5	N/A	N/A	- Selection - Information - Performance
Gros et al., 2017 United States	RCT Examine influence of mTBI on Prolonged Exposure to treat PTSD and SUD	N=51 (30 TBI+LOC; 21 Controls) TBI+LOC: - Sex [Males]: 97% - Age [Mean ± SD, Years]: 40.7 ± 11.1 - Race/Ethnicity: White (73%) - Marital Status: Single (33%); Married (17%) - Employment: Employed (36%) - Education [Mean ± SD, Years]: 13.6 ± 2.1 Controls: - Sex [Males]: 86% - Age [Mean ± SD, Years]: 38.7 ± 10.4 - Race/Ethnicity: White (62%) - Marital Status: Single (19%); Married (38%) - Employment: Employed (55%) - Education [Mean ± SD, Years]: 14.7 ± 2.1	mTBI/NR	SUD, PTSD/NR	Treatment	Prolonged Exposure (COPE - "Concurrent tx of PTSD and SUD using Prolonged Exposure")	Micro	Clinical	1.1	N/A	N/A	- Information - Selection - Performance - Ascertainment

Haarbauer-Krupa et al., 2017 United States	Cohort Examine incidence and factors associated w/PTSD 6 mos post-injury in civilian ED population	N=280 (75 w/PTSD; 205 w/out PTSD) w/PTSD: - Sex [Males]: 72% - Age [Mean ± SD, Year]: 42.0 ± 14.9 - Race: Caucasian (73%) - Marital Status: Single (59%); Married (20%); Separated/Divorced (16%) - Education [Mean ± SD, Year]: 13.5 ± 2.9 w/out PTSD: - Sex [Males]: 68% - Age [Mean ± SD, Year]: 43.3 ± 18.8 - Race: Caucasian (85%) - Marital Status: Single (51%); Married (35.1%); Separated/Divorced (6.3%) - Education [Mean ± SD, Year]: 14.7 ± 2.8	w/PTSD: Minor/Moderate (76%); Moderate/Severe/Critical (24%)/6 months w/out PTSD: Minor/Moderate (68%); Moderate/Severe/Critical (32%)/6 months	PTSD/NR	Screening	Screening for PTSD using PCL-C in ED	Meso	Clinical	4	N/A	N/A	N/A
Hart et al., 2020 United States	RCT Assess Behavioural Activation Therapy and SMS motivational messages to promote MHSU in individuals w/TBI	N=49 (38 Intentional Condition [INT]; 21 Motivation Condition [MOT]) INT: - Age [Mean ± SD, Years]: 40.4 ± 15.2 - Sex [Males]: 79% - Education [Mean ± SD, Years]: 13.2 ± 2.2 - Race/Ethnicity [White]: 44.7% MOT: - Age [Mean ± SD, Years]: 38.5 ± 15.3 - Sex [Males]: 81% - Education [Mean ± SD, Years]: 13.0 ± 1.8 - Race/Ethnicity [White]: 38%	INT: "At least complicated-mild injury severity"/ 7.4 ± 5.5 MOT: "At least complicated-mild injury severity"/ 4.4 ± 4.3	"At least mild depression and/or anxiety"/NR	Treatment	Behavioural Activation Therapy + SMS (text) messages: - Implementation intentions supporting individualized goals for increased rewarding/meaningful activities (8 weeks) Attention control session + SMS messages: - Motivational SMS messages (8 weeks)	Micro	Clinical	1.1 3.5	N/A	N/A	- Information - Selection - Performance - Ascertainment
Hofer et al., 2013 Switzerland	Case Study Describe psychotherapy and pharmacotherapy for OCD following TBI	N=1 - Gender: Man - Age: 27 years - Education: 12 years - Employment: Full-Time at time of injury	Severe/3 years	Daily marijuana use, daily cocaine use/Stopped consumption 1 year before TBI	Treatment	Psychotherapy (Prolonged Exposure, CBT) Pharmacotherapy (Paroxetine)	Micro	Service	1.1 4.1	N/A	N/A	- Selection - Information
Iliceto et al., 2018 United States	Case Study Demonstrate safe and successful use of rTMS for depression in patient w/hx of severe TBI	N=1 - Age [Years]: 37	Severe/NR	Anxiety, Bipolar/NR	Treatment	rTMS	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Performance - Reporting

Jak et al., 2019 United States	RCT Compared efficacy of CPT with SMART-CPT for comorbid PTSD and TBI	N=100 (49 CPT; 51 SMART-CPT) CPT: - Age [Mean ± SD, Years]: 33.9 ± 7.3 - Sex [Males]: 87.8% - Ethnicity: Caucasian (40.8%); African-American (16.3%); Asian (14.3%); Hispanic/Latino (20.4%); Native Hawaiian/Pacific Islander (4.1%); Native American/Native Alaskan (2.0%); Other (2.0%) SMART-CPT: - Age [Mean ± SD, Years]: 34.8 ± 8.5 - Sex [Males]: 90.2% - Ethnicity: Caucasian (52.9%); African-American (9.8%); Asian (3.9%); Hispanic/Latino (25.5%); Native Hawaiian/Pacific Islander (3.9%); Native American/Native Alaskan (2.0%), Other (2.0%)	CPT: Moderate (8.2%)/5.8 ± 4.0 SMART-CPT: Moderate (3.9%)/ 4.9 ± 3.0	PTSD/NR	Treatment	SMART-CPT: - All standard components and structures of CPT - Adaptations made to mTBI and cognitive complaints	Micro	Clinical	1.1	N/A	N/A	- Information - Performance - Attrition - Selection
Janak et al., 2017 United States	Before-After, No Control Investigate pre- to post-tx changes in PTSD and persistent post-concussive sxs	N=257 - Sex [Males]: 89% - Age: 18-24 years (29.6%); 25-29 years (22.2%); 30-37 years (26.0%); ≥38 years (22.2%)	Mild/Median = 5 months; IQR = 47-573 days	PTSD/NR	Program	Multidisciplinary Tx Program: - Psychotherapy, pharmacotherapy, vestibular rehabilitation, musculoskeletal therapy - OT, PT, medical personnel, psychologist, SLP	Micro	Service	1.1 4.1	N/A	N/A	- Selection - Information - Reporting - Performance
Jones et al., 2019 United States	Case study Describe military service personnel w/TBI and PTSD and describe participation in art therapy program	N=3 - Age [Years]: 23, 32, 31 - Gender [Men]: 100%	mTBI/NR	PTSD (N=3), Anxiety (N=2), Depression (N=1)/NR	Treatment	Stage-based art therapy	Micro	Clinical	1.1	- Empowerment (support from tx team, fellow group members, and community)	N/A	- Selection - Information - Performance - Reporting
Jorge et al., 2016 United States	RCT Assess efficacy of Sertraline in prevention depressive disorders following TBI	N=94 (48 Sertraline; 46 Placebo) Sertraline: - Sex [Males]: 54% - Age [Mean ± SD, Years]: 50.0 ± 20.1 - Race: White (98%) - Marital Status: Married (55%); Single (21%); Divorced (17%); Widowed (6%) - Employment: Employed (72%); Retired (17%); FT student (4%); Disabled (2%); Unemployed (2%) - Education [Mean ± SD, Years]: 13.5 ± 1.9 Placebo: - Sex [Males]: 65% - Age [Mean ± SD, Years]: 54.9 ± 18.2 - Race: White (98%) - Marital Status: Married (62%); Single (16%); Divorced (11%); Widowed (11%) - Employment: Employed (58%); Retired (33%); FT students (2%); Disabled (0%); Unemployed (2%) - Education [Mean ± SD, Years]: 14.1 ± 2.6	Sertraline: Mild (79%), Moderate (10%), Severe (10%)/NR Placebo: Mild (78%), Moderate (13%), Severe (9%)/NR	Mood Disorder	Treatment	Sertraline	Micro	Clinical	None	N/A	N/A	- Selection - Attrition

Juengst et al., 2015 United States	Before-After, No Control Describe development of smartphone application for assessing mood-related sxs after TBI and examine feasibility of using this in community-based sample of adults with TBI	N=20 - Sex [Males]: 60% - Age [Mean ± SD, Years]: 36.7 ± 12.4 - Race: 95% - Marital Status: Married (30%) - Education [Mean ± SD, Years]: 15.0 ± 2.3	"Complicated mild to severe TBI"/ 5.2 ± 3.6	Depression; Anxiety/NR	Screening	iPerform (user-centred cross-platform smartphone application) for self-monitoring of depression and anxiety sxs	Meso	Clinical	3.5	- Simplicity w/application interface and question content and style - Effective and efficient communication pathways b/t individuals and clinicians/researchers - Proper training	- Redundant and complex language - Alternating assessments (e.g., multiple scales for different questions, differing time frames of references for different questions) --> confusion	N/A
Jurick et al., 2020 United States	RCT Determine whether individuals w/scores in the invalid vs. valid range of the performance validity test show similar benefits from psychotherapy and if psychotherapy improves performance validity test performance	N=100 - Age (Mean ± SD, Years): 34.4 ± 7.9 - Sex [Males]: 89% - Race: Caucasian (70%) - Education (Mean ± SD, Years): 13.7 ± 1.8	Mild, moderate/5.4 ± 3.5	PTSD/NR	Treatment	SMART-CPT: - All standard components and structures of CPT - Adaptations made to mTBI and cognitive complaints	Micro	Clinical	1.1	N/A	N/A	- Information - Performance - Attrition - Selection
Kaimal et al., 2019 United States	Cross-sectional Present findings on participant feedback on short- and long-term art therapy service for military service members w/PTSD and TBI	N=204 Sex: N=204 Males N=12 Females (as reported in study)	N=57 Unknown; N=47 Mild; N=20 Moderate; N=1 Severe (as reported in study)/NR	PTSD/NR	Treatment	Long-term art therapy program in outpatient integrative medical care facility - Group and individual therapy	Meso	Clinical	1.1	N/A	N/A	- Information - Reporting - Selection - Low survey completion rate
Kip et al., 2019 United States	Before-After, No Control Present data for use of Accelerated Resolution Therapy among individuals with PTSD and TBI	N=202 - Age [Mean ± SD, Years]: 43.0 ± 12.8 - Sex [Males]: 90.1% - Race: White (83.6%); Black (12.9%); Other (3.5%) - Education [Mean ± SD, Years]: 14.5 ± 2.7	Mild (N=48); Moderate/Severe (N=49)/NR	PTSD/NR	Treatment	Accelerated Resolution Therapy: - Relaxation and orientation - Imaginal exposure - Imagery rescripting - Assessment and closeout	Micro	Clinical	1.1	N/A	N/A	- Selection - Performance

Levy et al., 2019 United States	Qualitative Examine the potential, usability, relevance, and acceptability of virtual reality grocery store as an assessment and intervention tool for veterans w/mTBI	N=6 focus groups - Therapist: N=6 at initial, N=7 at follow-up - Veterans w/mTBI and PTSD: N=3 at initial, N=4 at follow-up	mTBI/NR	PTSD/NR	Treatment	V-Mart - Prototype of an interactive, navigable, fully stocked, 7-aisle virtual grocery store designed as a therapist guided tool for assessment and tx of cognitive and emotional impairments associated with mTBI and/or PTSD	Meso	Clinical	1.1 3.5	N/A	For therapist: - Technical issues (difficult to use if not skilled w/game playing)	- Small sample
Luo et al., 2015 China	Cohort Investigate relationship between degree of depression in patients w/TBI and severity and progression of TBI; therapeutic effects of Prednisone w/psycho-interventions and anti-depression drugs	N=68 - Sex [Males]: 66% - Age [Years]: 18 to 70	All/NR	Depression/"observed from 3 mos - 3 years after TBI"	Treatment	Pharmacotherapy (Citalopram, Prednisone) Psychotherapy - Interpersonal therapy, behavioural activation, CBT, problem solving therapy, social skills training, psychodynamic therapy, supportive counseling	Micro	Service	1.1 4.1	N/A	N/A	- Selection - Performance - Information
Moss, 2018 United States	Case Study Describe integration of hypnosis and variety of palliative care strategies into multi-modal tx program	N=1 - Gender: Woman - Age [Years}: 38 - Marital Status: Married - Employment: Formerly employed	NR/2 years	Depression; PTSD/NR	Treatment	Hypnosis, energy therapy, psychotherapy, acupuncture, medication management	Micro	Service	1.1 4.1	N/A	N/A	- Selection - Information - Reporting - Performance
O'Connor et al., 2016 United States	RCT Evaluate 12 week cognitive rehabilitation intervention embedded w/vocational rehabilitation services for veterans w/hx of mTBI and MHSU	N=18 (10 cog rehab; 8 control) - Sex [Males]: 100% - Age [Mean ± SD, Years]: 51.0 ± 8.6 - Ethnicity: White (61.1%); Hispanic/Latino (22.2%); African-American/Black (16.7%) - Education [Mean ± SD, Year]: 13.0 ± 1.7; High School Diploma (83.3%); Associate's Degree (11.1%); Bachelor's Degree (5.6%)	mTBI/NR	PTSD (N=8); MDD (N=5); Bipolar (N=3); OCD (N=1); Schizoaffective Disorder (N=1); Generalized anxiety disorder; (N=1); Other anxiety and mood disorder; (N=7); Alcohol Abuse/Dependence (N=9); Opiate Abuse/Dependence (N=2); Cocaine Abuse (N=2) /NR	Treatment	Cognitive rehabilitation - Compensatory strategies to manage cognitive deficits - Laptop w/software to address training and rehabilitation needs - veteran + cognitive rehabilitation specialist met with vocational rehabilitation specialist	Meso	Service	1.1 3.5 4.1	- Lack of uniformity in module across exercises - Expanded word count (for computer response)	- Computer software (navigation that requiring restructuring of elements on screen, limited word count) - Remembering password	- Selection - Information - Performance - Ascertainment - Reporting

Olson-Madden et al., 2014 United States	Cohort Determine whether a positive screen on TBI-4 can be used to identify veterans who use more inpatient and outpatient MH services	AnyQ+/-: Any question positive/negative; Q2+/-: Question 2 positive/negative N=1,493 (AnyQ+: 942; AnyQ-: 551; Q2+: 657; Q2-: 836) AnyQ+: - Sex [Males]: 91.9% - Age [Mean ± SD, Years]: 47.6 ± 13.0 - Race/Ethnicity: White (74.5%); Black (11.7%); Hispanic (0.9%); Other (2.8%); Missing (10.2%) - Marital Status: Married (28.1%); Divorced/Separated/Widowed (46.8%); Single (24.2%); Missing/unknown (0.9%) AnyQ-: - Sex [Males]: 88.0% - Age [Mean ± SD, Years]: 47.8 ± 13.7 - Race/Ethnicity: White (70.4%); Black (17.2%); Hispanic (0.5%); Other (2.2%); Missing (9.6%) - Marital Status: Married (36.7%); Divorced/Separated/Widowed (39.9%); Single (23.2%); Missing/Unknown (0.2%) Q2+: - Sex [Males]: 94.7% - Age [Mean ± SD, Years]: 47.6 ± 13.1 - Race/Ethnicity: White (78.1%); Black (9.6%); Hispanic (0.6%); Other (2.0%); Missing (9.7%) - Marital Status: Married (28.9%); Divorced/Separated/Widowed (46.3%); Single (24.2%); Missing/Unknown (0.6%) Q2-: - Sex [Males]: 87.2% - Age [Mean ± SD, Years]: 47.8 ± 13.4 - Race/Ethnicity: White (69.0%); Black (17.0%); Hispanic (0.8%); Other (3.0%); Missing (10.2%) - Marital Status: Married (33.1%); Divorced/Separated/Widowed (42.7%); Single (23.6%); Missing/Unknown (0.6%)	NR/NR	NR (all patients sought MH services)/NR	Screening	Screening of TBI using TBI-4	Meso	Clinical	4	N/A	N/A	N/A
Ponsford et al., 2019 Australia	RCT Investigate factors associated w/positive response to adapted CBT intervention for individuals w/anxiety and depression following TBI	Motivational Interviewing (MI); Non-directive counseling (NDC) N=45 (22 MI + CBT; 23 NDC + CBT) MI+CBT: - Age [Mean ± SD, Years]: 47.7 ± 16.1 - Sex [Males]: 92.7% - Education [Mean ± SD, Years]: 13.7 ± 3.5 NDC+CBT: - Age [Mean ± SD, Years]: 40.4 ± 13.4 - Sex [Males]: 73.9% - Education [Mean ± SD, Years]: 12.9 ± 3.1	MI+CBT: GCS [Mean ± SD]; 10.7 ± 3.7/ 5.6 ± 12.3 NDC+CBT: GCS [Mean ± SD]; 10.2 ± 4.3/ 3.8 ± 6.2	Anxiety; Depression/NR	Treatment	CBT + Motivational interviewing (to enhance motivation and tx self-efficacy) CBT + Nondirective counseling CBT adapted for cognitive impairments	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Ascertainment - Performance
Ragsdale et al., 2016 United States	Cohort Study Examine effectiveness of completing prolonged exposure or CPT among veterans w/dx of PTSD with and w/out hx of TBI	N=41 (21 Prolonged Exposure; 20 CPT) - Sex [Males]: 87.8% - Age [Mean ± SD, Years]: 33.1 ± 8.2 - Race: Caucasian (85.4%); African American (7.3%); Hispanic/Latino (7.3%) - Marital Status: Never married (32.5%); Married (26.8%); Divorced (17.1%); Remarried (9.8%); Separated (9.8%); Living w/partner (2.4%); Unknown (2.4%)	NR/NR	PTSD/NR	Treatment	CPT Prolonged exposure therapy	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Performance - Reporting

Ragsdale et al., 2018 United States	Cross-sectional Compared the exposure therapy process in veterans w/PTSD alone to those w/PTSD and TBI hx	N=88 (45 TBI + PTSD; 43 PTSD only) TBI+PTSD: - Sex [Males]: 95.6% - Age [Mean ± SD, Years]: 36.3 ± 8.7 - Race: Caucasian (68.9%); African-American (15.6%); Hispanic (13.3%); Other (2.2%) - Education: High School (42.2%); Some College (31.1%); Bachelors (15.6%); Masters/Doctoral (11.1%) - Marital Status: Married (57.8%); Single (13.3%); Separated (17.8%); Divorced (11.1%) PTSD Only: - Sex [Males]: 93.0% - Age [Mean ± SD, Years]: 39.4 ± 9.7 - Race: Caucasian (67.4%); African-American (7.0%); Hispanic (14.0%); Asian (2.3%); Indian Subcontinent (2.3%); Other (7.0%) - Education: High School (23.3%); Some College (53.5%); Bachelors (20.9%); Masters/Doctoral (2.3%) - Marital Status: Married (46.5%); Single (20.9%); Separated (20.9%); Divorced (11.6%)	NR/NR	PTSD/NR	Treatment	Virtual reality assisted exposure therapy + group therapy	Micro	Clinical	1.1 3.5	N/A	N/A	- Information - Nondifferential misclassification - Selection - Performance
Rao et al., 2019 United States	RCT Acquire data to inform safety and tolerability of low-frequency right-sided rTMS in patients w/TBI and effective size over the dorsolateral prefrontal cortex on TBI depression and common comorbid psychiatric sxs	N=30 (17 Sham, 13 rTMS) rTMS: - Age [Mean ± SD, Years]: 39.8 ± 14.2 - Sex [Males]: 38% - Race [Caucasian]: 69% - Education [Mean ± SD, Years]: 14.6 ± 3.0 - Marital Status [Married/Partner]: 38% Sham: - Age [Mean ± SD, Years]: 40.2 ± 14.5 - Sex [Males]: 65% - Race [Caucasian]: 59% - Education [Mean ± SD, Years]: 14.4 ± 2.6 - Marital Status [Married/Partner]: 35%	rTMS: Mild (100%)/3 mos- 1 year (23%); 1-5years (38%); 5-10 years (23%); >10 years (15%) Sham: Mild (88%); Severe (12%)/ 3 mos-1 year (53%); 1-5 years (12%); 5-10 years (29%); >10 years (6%)	Shame: Depression/39.4 ± 43 rTMS: Depression/39.6 ± 36.1	Treatment	rTMS	Micro	Clinical	1.1	N/A	N/A	- Selection
Ravid et al., 2019 Israel	Case Study Present case study in which integrative approach combined psychiatric medicine and traditional Chinese medicine	N=1 - Gender: Woman - Age [Years]: 55 - Marital Status: Married - Employment: Previously employed	NR/1 year	Anxiety, patterns of avoidance, introversion, difficulty expressing herself, anhedonia (inability to feel pleasure) and difficulty interacting w/people/NR	Treatment	Pharmacotherapy, acupuncture, herbal formula	Micro	Service	1.1 4.1 4.3	N/A	- Financial limitations	- Selection - Information - Reporting - Performance

Roche et al., 2020 United Kingdom	Case Study Illustrate management of PTSD in context of TBI using Acceptance and Commitment Therapy-based approach	N=1 - Gender: Woman - Age [Years]: 48	Moderate/NR	PTSD/NR	Treatment	Acceptance and Commitment Therapy	Micro	Clinical	1.1	N/A	N/A	- Selection - Information
Schneider et al., 2016 United States	Cohort Determine whether a positive screen on TBI-4 is associated w/increased risk for suicide attempt w/in 1 year post-screening	N=1,097 (489 Positive; 629 Negative) Positive: - Sex [Males]: 94.4% - Age [Mean ± SD, Years]: 47.8 ± 12.9 - Race/ Ethnicity: Caucasian (76.9%); African American (10%); Hispanic (0.6%); Other (1.9%); Missing (10.5%) - Marital Status: Married (30.8%); Divorced/Separated/Widowed (45.5%); Single (22.9%); Missing/Unknown (0.9%) Negative: - Sex [Males]: 87.6 - Age [Mean ± SD, Years]: 48.2 ± 13.6 - Race/ Ethnicity: Caucasian (68%); African-American (17.2%); Hispanic (0.5%); Other (3.0%); Missing (11.3%) - Marital Status: Married (35.5%); Divorced/Separated/Widowed (40.7%); Single (23.1%); Missing/Unknown (0.8%)	NR/NR	NR (all patients sought MH services)/NR	Screening	Screening for TBI using TBI-4 by MH clinicians during MH intake	Meso	Clinical	4	N/A	N/A	N/A
Siddiqi et al., 2019 United States	Case Study Describe case in which resting state network mapping was successfully used to target rTMS to tx neuropsychiatric disturbances in retired NFL defensive linesman	N=1 - Gender: Man - Age: "Fourth decade of life"	NR/NR	Depression; Anxiety; Impulsivity; Anger/"2-3 year hx"	Treatment	rTMS	Micro	Clinical	1.1	N/A	N/A	- Selection - Information
Sripada et al., 2013 United States	Cohort (Study 1); RCT (Study 2) Assess impact of prolonged exposure for individuals with and w/out hx of TBI	Study 1: N=51 - Sex: NR - Age [Mean ± SD, Years]: 49.3 - Race: White (96%); Black (4%) Study 2: N=22 - Sex [Females]: 9% - Age [Mean ± SD, Years]: 32.7 ± 6.9 - Race: White (73%); African American (23%); Asian (5%)	mTBI/NR	Study 1: PTSD; SUD (17%); Comorbid Psychiatric (64%)/NR Study 2: PTSD; Depression (57%)/NR	Treatment	Prolonged exposure	Micro	Clinical	1.1	N/A	N/A	- Information - Nondifferential misclassification - Reporting - Ascertainment - Performance

Strom et al., 2016 United States	Case Study Demonstrate successful implementation of prolonged exposure in instances where the presence of TBI-related cognitive and behavioural deficits interfered w/standard tx implementation	N=2 - Gender: Man (N=1); Woman (N=1) - Age [Years]: 29 and 55 - Race/Ethnicity: Caucasian - Marital Status [Married]: 100%	Severe (N=1), moderate (N=1)/NR	PTSD/NR	Treatment	Prolonged exposure therapy - Compensatory strategies to account for cognitive deficits	Micro	Clinical	1.1	- Psychoeducation re: overlap of PTSD and TBI sxs - Protocol-consistent modifications to address cognitive and behavioural deficits associated with TBI	N/A	- Selection - Information - Performance
Walker et al., 2016 United States	Case Study Present therapeutic process for senior military service member who struggled w/untreated sxs of PTSD and moderate TBI	N=1 - Gender: Man - Age [Years]: "50s" - Marital Status: Married	Moderate/ "almost 7 years"	PTSD/NR	Treatment	Guided art therapy and acupuncture	Micro	Service	1.1	N/A	N/A	- Selection - Information - Reporting - Performance
Whiting et al., 2020 Australia	RCT Investigate if Acceptance and Commitment Therapy can facilitate psychological adjustment and reduce psychological stress following severe TBI	N=19 (10 ACT; 9 Control) ACT: - Age [Mean ± SD, Years]: 36.4 ± 13.5 - Sex [Males]: 80% - Education [Mean ± SD, Years]: 11.2 ± 2.0 Control: - Age [Mean ± SD, Years]: 37.2 ± 12.5 - Sex [Males]: 77.8% - Education [Mean ± SD, Years]: 11.4 ± 1.0	ACT: Severe/20.7 ± 17.5 Control: Severe/33.3 ± 21.5	Depression; Anxiety; Stress/NR	Treatment	Acceptance and Commitment Therapy - Strategies implemented to accommodate for cognitive impairment Befriending therapy (Control): - Therapy focuses on neutral topics that are unlikely to elicit a negative emotional response - Relationship with participant is friendly an engaging rather than empathic - Therapist provides positive statements rather than problem solving	Micro	Clinical	1.1	N/A	N/A	- Selection - Information
Winter et al., 2016 United States	RCT Test Veteran's In-Home Program to promote community re-integration, mitigate TBI sxs, and facilitate daily functioning	N=81 - Age [Mean ± SD, Year]: 40.1 ± 13.0 - Sex [Males]: Intervention (87.8%); Control (95%) - Race: White (58.0%); Black (35.8%); Native American (1.2%); Asian (1.2%); No Primary (1.2%); Other (2.5%) - Marital Status: Married (71.6%) - Employment: Employed (34.6%) - Education: <High School Degree (4.9%); 45.7% Some College Education (45.7%); 24.7% High School Degree/GED (24.7%); College Degree (16%); Post-Doc Degree (7.4%)	Intervention: Mild (70%)/ 9.98 ± 11.1 Control: Mild (58.3%)/ 9.48 ± 11.2	PTSD; Depression/NR	Program	Veteran's In-Home Program: - OT meets with veteran and family to identify concerns, review intervention goals, introduce strategies for goals, home modification	Meso	Clinical	1.1 4.1	- Family involvement --> supports tx adherence and maintenance and provides education and support to family	N/A	- Selection - Performance - Information

Wolf et al., 2018 United States	Before-After, No Control Examine generalization of the effectiveness of prolonged exposure therapy for PTSD in improving post-concussive sxs and other outcomes in military service members and veterans w/hx of mild to severe TBI	N=44 - Sex [Males]: 93.2% - Age [Mean ± SD, Years]: 33.2 ± 7.34 - Ethnicity: Caucasian (70.5%); Hispanic (18.2%); African American (9.1%); Other (2.3%) - Marital Status: Married (50%); Single (38.6%); Separated/Divorced (11.4%)	Mild (65.9%); Moderate/Severe (34.1%)/ 4.42 ± 2.99	PTSD; Depression (88.6%); Other Anxiety (27.3%); Substance Abuse (29.5%); hx of Suicide (29.5%)/ NR	Treatment	Prolonged exposure - Modifications to accommodate TBI-related deficits	Micro	Clinical	1.1	- Modifications to accommodate TBI-related deficits - Compensatory devices to provide support and corrective instruction - "teach-back" method to ensure understanding of content and instructions - recorded sessions and patient handouts	N/A	- Selection - Performance - Reporting
Yount et al., 2013 United States	Case Study Demonstrate effects of service dog training as purpose-driven intervention on PTSD sxs and mTBI	N=2 - Gender [Men]: 100%	NR/NR	PTSD/NR	Treatment	Training of service dogs for fellow veterans	Micro	Clinical	1.1	N/A	N/A	- Selection - Information - Reporting - Performance
Zhang et al., 2017 China	RCT Compare effect of methylphenidate w/placebo in patients with TBI in China	N=36 (18 Intervention; 18 Control) Intervention: - Age [Mean ± SD, Years]: 36.3 ± 10.9 - Sex [Males]: 72.2% - Ethnicity: Han (66.7%); Hui (33.3%) - Education: Elementary School (22.2%); High School (44.4%); College/University (33.3%) Control: - Age [Mean ± SD, Years]: 34.9 ± 12.1 - Sex [Males]: 77.8% - Ethnicity: Han (55.6%); Hui (44.4%) - Education: Elementary School (33.3%); High School (33.3%); College/University (33.3%)	Mild-moderate/NR	Major depressive episode/NR	Treatment	Methylphenidate	Micro	Clinical	1.1	N/A	N/A	- Selection - Information

ACT: Acceptance and Commitment Therapy; **ADHD:** Attention deficit hyperactivity disorder; **B/t:** between; **CBT:** Cognitive behavioural therapy; **CBT-T:** Cognitive behavioural therapy - telephone; **CBT-IP:** Cognitive behavioural therapy - in person; **CPT:** Cognitive processing therapy; **Dx:** Diagnosis; **ED:** Emergency department; **GCS:** Glasgow Coma Scale; **Gp:** Group; **HCP:** Healthcare professional; **Hx:** History; **IC:** Integrated care; **LOC:** Loss of consciousness; **LT:** Long-term; **MCBT:** Mindfulness based cognitive behavioural therapy; **MDD:** Major depressive disorder; **MH:** Mental health; **MHSU:** Mental health and/or substance use; **Mos:** Months; **mTBI:** Mild traumatic brain injury; **N/A:** Not applicable; **NR:** Not reported; **OCD:** Obsessive compulsive disorder; **OSU-OT:** Occupational therapy; **TBI-ID:** Ohio State University Traumatic Brain Injury Identification Method; **PT:** Physiotherapy; **PTSD:** Post-traumatic stress disorder; **QoL:** Quality of life; **rTMS:** Repetitive transcranial magnetic stimulation; **RCT:** Randomized controlled trials; **SD:** Standard deviation; **SLP:** Speech language pathology; **SPT:** Supportive psychotherapy; **SUD:** Substance use disorder; **Sxs:** Symptoms; **TBI:** Traumatic brain injury; **UC:** Usual care; **Tx:** Treatment**; W/:** With